Workload-Aware Digitally Assisted Fast-Transient Capacitorless LDO with Adaptive Biasing for Edge-AI Systems in SCL 1.2 µm CMOS

Revised faculty-approval proposal | Analog/Mixed-Signal IC Design

1. Project Identification

Field	Revised specification
Project Title	Workload-Aware Digitally Assisted Fast-Transient Capacitorless LDO with Adaptive Biasing for Edge-AI Systems in SCL 1.2 µm CMOS
Category	Analog / Mixed-Signal IC Design
Target Technology	SCL Punjab 1.2 µm CMOS PDK
Primary Application	Local power regulation for bursty Edge-AI / AIoT processing domains
Research Focus	Fast transient response + low standby power without a large external output capacitor

1.1 Team & Mentorship

Jonathan Gerard Joseph (24BVD0198), Karthik Kandimalla (24BVD0094), Mosangi Viswaj (24BVD0221), Adinath Bharathan (24BVD0152), Vidit Narayan Panigrahi (24BVD0167), Panguluri Dattasai Bhargav (25BVD0103), Rishabh Gupta (25BVD0098), Kattamuri Raghavendra (25BVD0124).

Primary Mentor: Dr. Jyotishman Saikia, Department of Micro and Nanoelectronics, SENSE

Co-Mentor: Dr. Vaegae Naveen Kumar, Department of Communication Engineering

2. Technical Proposal

2.1 Abstract

Edge-AI and AIoT devices usually work in different modes. Sometimes they stay in a low-power state for sensing or standby, and sometimes they suddenly need more power for computing, communication, or AI processing. These sudden changes in workload can cause quick changes in the current, so the device needs a power supply that can respond quickly.

In this project, we are designing an Output-Capacitorless Low-Dropout Regulator (OCL-LDO) using the SCL Punjab 1.2 µm CMOS process. The main advantage of an OCL-LDO is that it can provide a stable voltage without using a large external capacitor. However, designing a fast and stable LDO with low power consumption is challenging in a 1.2 µm CMOS process.

Our design uses an FVF regulation loop, a transient-boosting circuit, adaptive biasing, and a 4-bit digital voltage-control system. The adaptive biasing increases the drive when the load suddenly changes, helping the LDO respond faster. A digital transient detector will also help detect sudden changes and improve the response.

If possible, an AI controller will provide an early signal before a heavy workload starts. This allows the LDO to prepare for the upcoming increase in power demand. The main aim is to improve the LDO's response and energy efficiency while maintaining good voltage regulation, stability, PSRR, and reasonable chip area

2.2 Final Application - in simple terms

- The LDO we are designing is not an AI accelerator. It is a small voltage regulator that works along with an Edge-AI processor, AIoT device, sensor hub, DSP, or other accelerator. Its main job is to take an unstable power supply and provide a clean and stable voltage to the circuit. It can also respond quickly when the device suddenly needs more power.
- For low-power Edge-AI sensor devices, the LDO can help when the device changes between sleep, sensing, AI processing, and communication.
- For Edge-AI cameras and agricultural or industrial devices, it can handle sudden increases in power when the device processes images or signals.
- In mixed-signal chips, it can help protect sensitive sensor and analog circuits from power-supply noise.
- For AIoT devices, dynamic voltage scaling can help reduce power consumption. When the device suddenly wakes up or starts AI processing, the LDO helps keep the power supply stable.

2.3 Research Hypothesis & Objectives

Hypothesis: In a 1.2 µm CMOS process, using adaptive biasing and a fast digital transient path can help the LDO respond faster to sudden load changes and improve energy efficiency without needing a large external capacitor.

- First, We will study the SCL 1.2 µm devices and understand their characteristics before deciding the final specifications.
- We will design a basic FVF OCL-LDO and use it as a reference for comparison.
- We will add adaptive biasing so that the circuit can use more current when the load changes suddenly.
- We will develop a digital transient detector and boost circuit to quickly control the pass device.
- We will keep the 4-bit DVS interface and choose its voltage range based on the process and application requirements.
- We will also study a workload-hint signal that can prepare the LDO before an expected AI processing or wake-up event.

- Finally, We will check the stability, voltage regulation, PSRR, transient response, and PVT performance, followed by layout, PEX, AMS verification, and silicon testing if fabrication is available.

2.4 Proposed Architecture

- **FVF and pass element: This part controls the output voltage and helps keep it stable.**
    - **Dynamic transient-boost path: It detects sudden changes in the output voltage and quickly gives extra drive to the pass device. This helps the LDO respond faster.**
    - **Adaptive bias and stability network: It changes the bias current depending on the load. This helps maintain a good balance between response speed, stability, and power consumption.**
    - **4-bit voltage control/DVS: This allows us to change the output voltage using digital control. A workload-hint signal can also prepare the LDO before a sudden increase in load.**

For the initial design, we will consider a **0-**10 mA load range. We will only increase this range if the SCL PDK, pass device, chip area, and reliability allow it.

2.5 Research Innovation

- We will design the LDO specifically for the SCL 1.2 µm process instead of directly using a design made for a smaller technology.
- We will compare fixed-bias, adaptive-bias, and assisted designs to see how much power can be saved.
- We can use a workload signal to tell the LDO when an AI task is about to start. This allows the LDO to prepare before the load increases.
- We will test the LDO using different current patterns similar to Edge-AI workloads.
- We will include DVS controls and some test points so that the chip can be tested more easily after fabrication.

2.6 Design & Verification Flow

- **Cadence Virtuoso/Spectre: We will use it to design the LDO and check its performance through different simulations.**
- **SystemVerilog/QuestaSim: We will use it to test the 4-bit DVS and digital control logic.**
- **AMS co-simulation: We will check whether the analog LDO and digital controls work properly together.**
- **Layout + PEX: We will create the chip layout and check the effects of unwanted resistance and capacitance.**
- **DRC/LVS/ERC and GDSII: We will perform the final design and layout checks before fabrication.**
- **After fabrication: If the chip is fabricated, we will test it using a PCB, power supply, oscilloscope, and electronic load.**

Category	Metrics	Why
DC	Load regulation, line regulation, dropout	Basic regulation quality
Power	IQ, peak current, efficiency, energy/transient	Low-power suitability
Transient	Undershoot, overshoot, recovery/settling time	Primary research metric
AC	Loop gain, phase margin, PSRR, output noise	Stability and mixed-signal suitability
Digital	DVS monotonicity, code-to-output mapping, control latency	Control correctness
Physical	Area, PEX impact, routing/parasitics	Mature-node feasibility
Application	Response to sleep→sensing→inference →communication profiles	Edge-AI relevance

2.8 Expected Outcomes

- We will design a basic capacitorless LDO and an improved LDO using the SCL 1.2 µm CMOS process.
- We will compare the different versions of the LDO to find which design gives better performance and uses less power.
- We will test the LDO with different Edge-AI workload patterns and check how it handles sudden changes in load.
- We will complete the chip layout and check it using DRC, LVS, PEX, and AMS verification. If possible, we will prepare the final GDSII file.
- If chip fabrication is available, we will test the fabricated chip. If fabrication is not available, we will complete the project using post-layout simulations and verification.

3. One-Year Execution Plan

Phase	Period	Activities	Leads
1	Aug–Sep 2026	Literature, SCL PDK/device characterization, specifications	All
2	Oct–Dec 2026	Baseline FVF + adaptive bias + transient assist + DVS	Analog & Digital tracks
3	Jan–Mar 2027	AMS/PVT/stability/PSRR/workload optimization	All
4	Apr–May 2027	Custom layout, integration, PEX	Analog + Digital
5	Jun–Jul 2027	DRC/LVS/ERC, AMS sign-off, GDSII/tape-out	All
6*	Aug 2027 onward	Fabrication return, PCB bring-up,	All

		silicon characterization, paper/report	

The fabrication and silicon testing will happen only if the MPW schedule allows it.

4. Literature Position

Previous studies have already worked on transient detection in capacitorless FVF-LDOs and adaptive biasing in OCL-LDOs. Some recent studies have also focused on dynamic current control, feed-forward control, and LDOs for AIoT applications. Edge-AI systems are also using workload information to manage power and voltage.

So, we cannot say that these individual techniques are completely new. Instead, our main idea is to **combine these techniques and study their performance together** in the SCL 1.2 µm CMOS process. We will mainly focus on workloads where the power demand changes suddenly, as seen in Edge-AI devices.

5. Conclusion

This proposal presents a workload-aware digitally assisted capacitorless LDO regulator designed specifically for the SCL Punjab 1.2 µm CMOS process. By combining adaptive biasing, fast transient enhancement, and programmable voltage scaling, the design aims to achieve superior transient performance, low standby power consumption, and stable operation for modern Edge-AI applications.